Launcher Test

Does the flywheel spin the right way, where should the trigger gate sit, when is the wheel READY, and do shots land? (issue #5, decisions 029, 032)

Names		Date	
Battery voltage at start		Battery at end	

OpMode: Launcher Test (Tests group on the Driver Station). **Needs in the config:** `launcherMotor` (Motor, exact goBILDA type), `launcherTriggerServo` (Servo)

Safety: robot on blocks or on the floor with space around it, hands clear of moving parts, one person ready to press STOP. Nothing moves during INIT; everything starts after START.

The flywheel is fast and the ball comes out hard. Nobody in front of the launcher. Aim at a wall or a net for steps 1 to 5. Eye protection on.

Controls (gamepad 1)

Button	Does
A	start / stop the flywheel at the test power
B	STOP the flywheel
Dpad UP / DOWN	test power +10% / -10%
X / Y	test power 50% / 100%
HOLD right trigger	power follows the trigger (0 to 100%)
Dpad LEFT / RIGHT	flip the motor direction (for step 1)
HOLD left bumper	trigger gate to FIRE (open). Let go = REST (closed)

Steps, in order

#	Do this	Expected	Actual (what happened)	If wrong
1	X, then A	wheel spins so a ball would go OUT the launcher		flip with dpad; if FORWARD is right, `MOTOR_REVERSED = false`
2	Wheel stopped. Put a ball in the path, let go of the left bumper	gate at REST BLOCKS the ball		change `TRIGGER_REST_POSITION`
3	Hold the left bumper	gate at FIRE lets the ball pass freely		change `TRIGGER_FIRE_POSITION`
4	Y, then A. Wait 3 s. Read "Launcher Velocity"	a steady number: _____ ticks/s		`READY_VELOCITY_THRESHOLD` = about 90% of it: _____
5	B, then A again. Count seconds until "READY"	under 2 s		note it; the operator must wait this long

#	Do this	Expected	Actual (what happened)	If wrong
6	Launch one ball (A, wait READY, hold left bumper)	ball comes out		jam: check the gate positions and the water wheel

Power vs distance

Put the robot a set distance from the goal. For each power, launch 5 balls. The hit-rate test at the end uses the best power.

Power	Distance to goal (in)	Made / 5	Too short / too long?	Notes
60%				
80%				
100%				
___%				

Hit-rate test (the "done when" of issue #5)

Power	Distance (in)	Battery (V)	Made / 10	Who launched

Numbers to put into the code: `subsystems/Launcher.java`

Constant	Was	New value	Checked by
`MOTOR_REVERSED`	true		
`TRIGGER_REST_POSITION`	0.0		
`TRIGGER_FIRE_POSITION`	0.5		
`READY_VELOCITY_THRESHOLD`	1000		
`DEFAULT_SPIN_POWER`	1.0		

Also write in `hardware-config.md`: motor gear ratio _____, Expansion Hub motor port ____, trigger servo hub ____ port ____.

After the test: a coder puts the new numbers into the constants and removes the matching TODO comments; the results go on GitHub issue #5. Photos of anything surprising!